

Assessment of flow stability boundaries in a heated channel with different fluids at supercritical pressure

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ABSTRACT

This paper presents results on the stability of a simple heated channel containing fluids at supercritical pressure with an external imposed pressure drop. Basing on a recent work that discussed stability characteristics in a fluid-to-fluid comparison perspective, additional data are presented in order to discuss relevant parametric effects, also including a more accurate fluid-to-fluid comparison.

Three different analysis tools, including a system code and in-house linear and transient analysis programs, were adopted to evaluate stability thresholds at different channel throttling conditions and orientations; four different supercritical fluids were considered. The diversity of the adopted tools and the good level of agreement observed in the comparison of their results provide adequate confidence on the general reliability of the obtained information.

Among the addressed phenomena, both Ledinegg and density-wave oscillations are considered, pointing out a fundamental continuity between the two phenomena that occur in adjoining regions of the parameter space. Numerical effects are also highlighted, quantifying the impact of truncation error occurring in the use of system codes in the analysis of flow instabilities. The results obtained by different fluids provide further support to the conclusion that, when an appropriate dimensionless formalism is used, the differences obtained in the stability behaviour at imposed heat flux are relatively small for a number of fluids of interest for experimental analyses.

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1. Introduction

A basic understanding of the fundamental phenomena occurring when heated channels operate in conditions close to flow stability thresholds is necessary for addressing the complex behaviour occurring in nuclear reactor cores where the fluid undergoes considerable expansion. As known, this situation is common to subcritical and supercritical pressure systems when the saturation or the pseudo-critical temperature values are crossed by the bulk fluid along the channel, giving rise to considerable axial density gradients. The occurrence of an apparently negative internal pressure drop characteristic, either in static conditions or at some frequency of the inlet flow perturbation, results in flow excursions or self-sustained oscillations, recognised respectively as the classical Ledinegg or density-wave oscillation instability mechanisms (see e.g., Lahey and Moody, 1993).

In past decades, these phenomena were investigated at a considerable level of detail and complexity in the case of boiling channels. The related understanding was summarised in works appeared at different steps of the researches (Bourè et al., 1973;

March-Leuba and Rey, 1993; D'Auria et al., 1997); these works testified for an increasing insight into the related physical details and for the production of codes and models capable to address even full core simulations with a considerable degree of realism. This useful background of knowledge and simulation tools is available to be used for the case of reactor cores cooled by supercritical water, that are predicted to be possibly susceptible to similar instability mechanisms.

As known, at supercritical pressures fluids show remarkable changes in properties in a narrow temperature range across the “pseudo-critical” conditions, being defined by the temperature at which the specific heat reaches a sharp maximum at a given pressure above the critical one. Besides sharp changes observed in transport properties, density also shows a steep change, similar to that observed in boiling systems as a consequence of voiding. This is the main reason for expecting that instability effects similar to the ones observed in boiling systems may also occur at supercritical pressures.

The issue of stability is therefore of great importance in the design of the proposed Generation IV Supercritical Water Reactors (SCWRs) as pointed out in works devoted to address this issue in proposed plant configurations (see, e.g., Yi et al., 2004a,b; Churkin et al., 2010). In similarity with the case of Boiling Water Reactors

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Nomenclature

Roman letters

A	cross section area [m^2]
C_p	specific heat at constant pressure [$\text{J}/(\text{kg K})$]
$C_{\max}$	maximum Courant number
D_h	hydraulic diameter [m]
f	friction factor
$f_q(z)$	normalized distribution of heat flux
Fr	Froude number
g	gravity [m/s^2]
G	junction mass flux [$\text{kg}/(\text{m}^2\text{s})$]
h	fluid specific enthalpy [J/kg]
j	volumetric flux [m/s]
K_{in}	localized pressure drop coefficient at the channel inlet
K_{out}	localized pressure drop coefficient at the channel outlet
L	channel length [m]
N_{SPC}	sub-pseudo-critical number
N'_{TPC}, N_{TPC}	apparent and true trans-pseudo-critical numbers
p	pressure [Pa]
q''	heat flux [W/m^2]
t	time [s]
T	temperature [K] or period of oscillations [s]
T_{trans}	transit time along the channel = $\int_0^L \frac{dx}{w}$
v	specific volume [m^3/kg]
w	velocity [m/s]
z	axial coordinate along the channel [m]

z_I	imaginary part of a complex exponent
z_R	real part of a complex exponent

Greek letters

β	isobaric thermal expansion coefficient [K^{-1}]
δ_d	dimensional Dirac delta function [m^{-1}]
δ^*	dimensionless Dirac delta function
Λ	friction dimensionless group (Euler number)
Π_h	heated perimeter [m]
ρ	density [kg/m^3]

Subscripts

f	saturated liquid
g	saturated vapour (steam)
fg	difference between saturated vapour and saturated liquid
in	inlet
out	outlet
p	constant pressure
pc	pseudo-critical
0	reference value

Superscript

*	starred variables indicate dimensionless values
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(BWRs), it can be foreseen that the interplay of intrinsic thermal-hydraulic instabilities with the neutronic feedback might lead to possible unstable conditions to be carefully avoided in plant operation. Appropriate models must be therefore developed for predicting the fundamental flow phenomena on the basis of the experience gained in the field of BWR stability, waiting for applicable experimental data to come in support to model development as the presently available experimental information is scarce (see the extensive account on this aspect reported in Chapter 14 of the textbook by [Pioro and Duffey \(2007\)](#)).

In order to update the predictive tools for the application to supercritical water reactors, it is useful to try benchmarking them in a code-to-code comparison perspective. In this aim, reverting to the simple situation of the single heated channel with well defined imposed boundary conditions is helpful to reflect on the basic instability mechanisms operating in supercritical pressure flow conditions and on their possible differences and similarities with respect to the case of boiling channels.

Several papers were recently published in this regard addressing simple systems, in order to acquire a greater understanding of the related mechanisms (see e.g., [Chatoorgoon 2001, 2006](#); [Chatoorgoon et al., 2007](#); [Jain and Rizwan-uddin, 2008](#); [Ortega Gómez et al., 2008](#); [Rohde and van der Hagen, 2009](#); [Zhao et al., 2005](#)). Among the issues discussed in these papers, the following are given particular attention:

- static and dynamic nature of the instabilities predicted to be observed in systems with fluids at supercritical pressure;
- dimensionless parameters to be adopted in a dynamic similarity perspective to propose stability thresholds from calculations and future experiments in a compact and relatively universal way;
- effects related to single and multiple parallel channels as well as to systems in natural circulation.

Though a complete consensus on some of these aspects is not yet reached, as it is expected when different proposals are made

for dealing with new situations, a convergence towards common views should be not too far. This objective is one of the aims of ongoing international projects in the field that also involve the comparison between the predictions of different models and tools in order to get consensus at least on the basic modelling assumptions, while waiting for relevant experimental tests to say a final word about their adequacy.

In this respect, the work proposed in this paper represents a further step of an attempt to make use of previous concepts applicable to boiling channels in the case of fluids at supercritical pressure. Previously accomplished steps in this research can be synthesised as follows:

- particular definitions of dimensionless parameters were identified providing a rather surprising degree of generality to conclusions obtained by dimensionless equations on static and dynamic instabilities at imposed heat flux, nearly irrespectively of the value of the supercritical pressure and of the adopted fluid ([Ambrosini and Sharabi, 2006, 2008](#));
- using these definitions, many similarities can be highlighted between boiling and supercritical fluid instabilities in heated channels, resulting in a very intuitive correspondence between phenomena observed in the former and the latter systems ([Ambrosini, 2007](#));
- an independent assessment of the conclusions obtained by in-house models making use of the above dimensionless definitions was achieved by adopting system codes and comparing their results, obtained in the space of practically feasible operating conditions, with the hypothetical ones provided in dimensional form ([Ambrosini and Sharabi, 2007](#));
- unlike in the case of boiling channels, available CFD codes were found to be enough well developed to be immediately used to address stability phenomena in heated channels with supercritical water both in axi-symmetric and in 3D geometrical conditions ([Sharabi et al., 2008, 2009](#)); the substantial agreement observed between the phenomena shown by 1D models and CFD codes, to be considered as very diverse computational tools,

strongly supports the idea that system codes might be able to grasp the fundamental features of the considered instability mechanisms, something to be confirmed only by comparison with experiments.

In these developments, a specific effort is needed to demonstrate the assumption that the proposed dimensionless parameters are general enough to be applicable to different fluids, at least in the assumption of imposed channel heating that rules out possible differences in the heat input provided to the fluid as a consequence of the difference in heat transport properties. This assumption was somehow evident from the very beginning considering that the adopted dimensionless definitions provided similar trends of dimensionless density as a function of dimensionless enthalpy for different fluids; in the light of this observation, based on the experimental data on which current fluid property packages are developed (see, e.g., NIST, 2002), it was clear from the very beginning that such an assumption had enough sound bases to be at least proposed to the scientific community for the due independent assessment. On the other hand, whenever fully confirmed, this assumption could have an impact on the rationale for selecting fluids for experimental facilities, making eligible for the study of instabilities at imposed heat flux a broader class of fluids than presently believed possible.

A first computational confirmation of this assumption came from an unpublished work performed at the University of Pisa in cooperation with Tsinghua University (Zhang, 2009) analysing by a CFD code carbon dioxide in conditions that, in dimensionless form, were similar to those addressed by Sharabi et al. (2008) with water; the good match obtained between the observed stability thresholds reassured that the assumption could be more accurately confirmed by further investigation.

Recently, further results of the investigation were published (Ambrosini, 2009) making use of a purposely developed in-house transient code adopting the properties of four different fluids (water, carbon dioxide, ammonia and refrigerant R23); these results showed a qualitative agreement with the stability maps obtained in dimensionless form by a linear stability program: in particular, it was shown that excursive or oscillatory behaviours were observed exactly at the same inlet conditions at which they were predicted by the maps.

In the present paper, a more convincing presentation of similar results, obtained for a slightly different system, will be presented. In particular, after an accurate comparison of RELAP5 predictions for water at 25 MPa at different outlet throttling with the stability boundaries obtained in dimensionless form by the in-house linear stability code, the fluid-to-fluid comparison will be given greater attention considering the same fluids addressed in the previous work. Other aspects that will be given attention are:

- the effect of vertical and horizontal channel orientations;
- the problem of numerical diffusion and of its impact on the obtained stability analysis;
- the effect of channel pressure drop on stability;
- the continuity existing between the static and dynamic instability regions in both boiling and supercritical pressure systems.

2. Addressed system and boundary conditions

The addressed geometrical configuration is represented in Fig. 1. As in Ambrosini (2009), it consists of a circular pipe with uniform heating along the axis. The length is assumed to be 4.2672 m (14 ft) and the ID is 8.36 mm. This diameter was selected in a previous work (Sharabi et al., 2008) in order to keep the same cross section and the same length of a previously addressed SCWR subchannel, though the geometry was simplified in order to allow

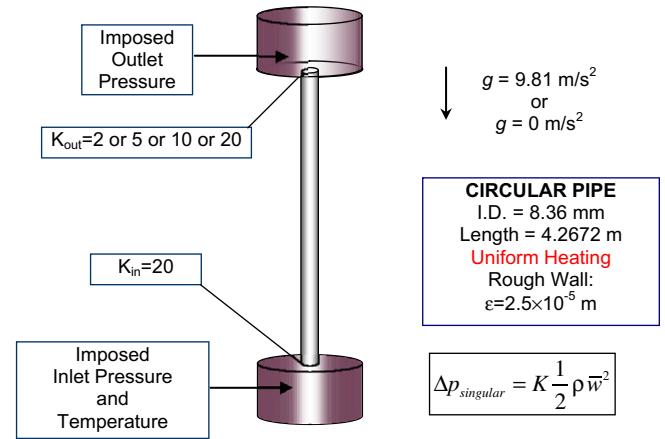


Fig. 1. Sketch of the addressed system.

for 2D axi-symmetric CFD calculations, performed in previous works (Sharabi et al., 2008; Zhang, 2009).

Inlet and outlet pressure drops represented by (irreversible) singular pressure drop coefficients ($K_{in} = 20$ and K_{out} variable from 2 to 20) are imposed and a uniform value of the roughness parameter, $\varepsilon = 2.5 \times 10^{-5}$ m, is assigned for the pipe wall; in this respect, the considered system is different from the one proposed in Ambrosini (2009). The existence of inlet and outlet plena of comparatively very large area is assumed at both sides of the channel. Both vertical upward flow and horizontal channel are assumed in the calculations.

As in the above mentioned work, the addressed fluids are those reported with their pseudo-critical properties in Table 1. Inlet temperature conditions for water and the other different fluids are specified making use of the sub-pseudo-critical number defined as:

$$N_{SPC} = \frac{\beta_{pc}}{C_{p,pc}} (h_{pc} - h_{in}) \quad (1)$$

In particular, values of N_{SPC} from 0.5 to 3 in steps of 0.5 are considered for water, while for the other fluids smaller sets of calculation cases were considered. In order to represent different heat input conditions, the trans-pseudo-critical number was instead used:

$$N_{TPC} = \frac{\dot{Q}_{heating}}{W_{in}} \frac{\beta_{pc}}{C_{p,pc}} \quad (2)$$

This allows expressing the stability threshold for any addressed condition in terms of a threshold value of N_{TPC} as a function of N_{SPC} .

The proposed problem must be regarded as an investigation about stability thresholds involving conditions that may be reached or not in practical applications. It is in fact predicted that instabilities can be observed only at relatively large power-to-flow ratios, involving in some cases high fluid temperatures, that should be hopefully far from normal operating conditions. In this respect, it must be explicitly noted that some of the identified unstable operating conditions could be predicted in dimensional form only owing to the forgiving nature of the adopted codes, that allowed to calculate even very high fluid temperature conditions without excessive numerical difficulties.

In relation to the particular choice for the adopted dimensionless formalism, it is here recalled that N_{SPC} and N_{TPC} represent the supercritical pressure counterpart of the subcooling and the phase-change numbers classically adopted for boiling systems in the so-called Ishii-Zuber plane (as shown in Ambrosini and Sharabi (2006, 2008)); a relevant difference is that the pseudo-critical

Table 1

Pseudocritical parameters evaluated according to NIST for the selected fluids at the considered operating pressures.

Fluid	p [MPa]	T_{pc} [K]	ρ_{pc} [kg/m ³]	h_{pc} [J/kg]	β_{pc} [1/K]	$C_{p,pc}$ [J/(kg K)]
Water	25.0	658.04	317.03	21,52,539	0.129	76,445
CO ₂	8.0	307.82	459.49	341,448	0.299	35,267
R23	5.7	306.37	537.01	288,279	0.127	9941
Ammonia	15.0	422.55	239.29	13,14,996	0.055	24,292

temperature now takes the place of the saturation range in defining the reference conditions. In this respect, many of the properties possessed by the subcooling and the phase-change numbers are also possessed by N_{SPC} and N_{TPC} , though the different equations of state and other details of the adopted definitions do not allow to immediately translate practical stability criteria applicable to boiling systems into corresponding ones for supercritical pressure systems. An account of similarities and differences between stability phenomena predicted for boiling and supercritical pressure systems is reported in Ambrosini (2007).

3. Adopted models

As in previous published work (Ambrosini, 2009), the adopted codes and models are:

- an in-house code making use of linearised and dimensionless forms of the space and time discretised governing equations, adopted to set up stability maps;
- an in-house transient code, used for dealing with the cases of different fluids in dimensional form (TRANSDIM);
- the RELAP5/MOD3.3 code (SCIENTECH Inc., 1999), used in transient mode only in the case of water.

All these models have in common the assumption of 1D flow, obtained by space and time averaging the local and instantaneous balance equations for mass, momentum and energy. The balance equations are therefore similar, though the diversity of the adopted tools makes the obtained results relatively independent from each other. The use of the 1D formulation represents the main limitation in the scope of the present work; nevertheless, it must be considered that the purpose is here to provide reference data for the discussion of the behaviour of system codes, having mostly 1D capabilities. The comparison of the results obtained by 1D models with those produced by 3D CFD models has been the subject of previous work (Sharabi et al., 2008, 2009).

3.1. Linear stability model

The linearised program (named ISHII-SUP) is based on dimensionless definitions introduced in previous works (Ambrosini and Sharabi, 2006, 2008) in which the model is described in detail. Here it is only recalled that the dimensionless definitions are as follows:

$$\rho^* = \frac{\rho}{\rho_{pc}} \quad h^* = \frac{\beta_{pc}}{C_{p,pc}}(h - h_{pc}) \quad w^* = \frac{w}{w_0} \quad t^* = \frac{tw_0}{L} \quad (3)$$

$$z^* = \frac{z}{L} \quad p^* = \frac{p}{\rho_{pc} w_0^2} \quad G^* = \rho^* w^* \quad Fr = \frac{w_0^2}{gL} \quad (4)$$

$$A = \frac{fL}{2D_h} \quad N'_{TPC} = \frac{q_0'' \Pi_h L}{\rho_{pc} w_0 A} \frac{\beta_{pc}}{C_{p,pc}} \quad N_{SPC} = \frac{\beta_{pc}}{C_{p,pc}}(h_{pc} - h_{in}) \quad (5)$$

$$N_{TPC} = \frac{\dot{Q}_{channel}}{W_{channel}} \frac{\beta_{pc}}{C_{p,pc}} = N'_{TPC} / \rho_{in}^* \quad (6)$$

The dimensionless mass, momentum and energy balance equations on which the model is based are:

$$\frac{\partial \rho^*}{\partial t^*} + \frac{\partial \rho^* w^*}{\partial z^*} = 0 \quad (7)$$

$$\begin{aligned} \frac{\partial G^*}{\partial t^*} + \frac{\partial}{\partial z^*} \left(\frac{G^{*2}}{\rho^*} \right) + \frac{\partial p^*}{\partial z^*} \\ = -\frac{\rho^*}{Fr} - \left[A + \frac{1}{2} K_{in} \delta^*(z^*) + \frac{1}{2} K_{out} \delta^*(z^* - 1) \right] \frac{G^{*2}}{\rho^*} \end{aligned} \quad (8)$$

$$\frac{\partial \rho^* h^*}{\partial t^*} + \frac{\partial \rho^* w^* h^*}{\partial z^*} = N'_{TPC} f_q(z^*) \quad (9)$$

In the development of these equations, it was assumed that:

- the flow is one-dimensional;
- the fluid expands because of heating, but it can be considered incompressible;
- the flow work can be neglected;
- heating because of viscous dissipation can be neglected.

The boundary conditions imposed to evaluate the steady-state conditions whose stability is the objective of the study are:

$$h_{in}^* = -N_{SPC} \quad (10)$$

$$G^* = \rho_{in}^* \quad (11)$$

The latter condition follows from the fact that w_0 is assumed to be the inlet velocity, w_{in} , so that $w_{in}^* = 1$ and $G_{in}^* = \rho_{in}^* w_{in}^* = \rho_{in}^*$. As a consequence of these boundary conditions, also considering that a unique relationship $\rho^*(h^*)$ is applicable for different supercritical pressures and fluids, given any set of parameters A , K_{in} , K_{out} , Fr , N_{SPC} and N_{TPC} the overall dimensionless pressure drop across the channel in steady-state conditions, $\Delta p_{channel}^*$, is completely specified. Also the distribution of dimensionless fluid density along the channel will be therefore completely specified since the dimensionless enthalpy distribution is specified.

In the stability analyses, the value of the pressure drop obtained by the steady-state calculation is imposed across the channel, letting flow rate to freely oscillate. The dynamics of the system is in fact fully defined and the threshold value of N_{TPC} leading to unstable behaviour can be identified as a function of N_{SPC} , when all the other parameters are assumed constant.

The above balance equations are discretised in space and time by a classical semi-implicit numerical scheme, in similarity with what made in previous work related to boiling channels (Ambrosini et al., 2000). The linearisation by perturbation of the obtained algebraic equations is followed by the evaluation of the eigenvalues of a matrix representing at each steady-state condition the linearised dynamics of the system, providing the following parameters:

- z_R , being a dimensionless amplification parameter that is negative for stable conditions and positive for unstable ones;
- z_i , being a dimensionless angular frequency of oscillations.

While z_R is used to set up contour plots used as quantitative stability maps, allowing for identifying the stability boundary in the $N_{SPC} - N_{TPC}$ plane, z_I is used to evaluate the dimensionless period of oscillations that can be conveniently compared with the dimensionless transit time, producing interesting contour plots of their ratio.

The discretisation method adopted for the balance equations, making use of staggered meshes and of a classical donor cell principle (upwind differencing), has a first order truncation error. A reasonable compromise between computational efficiency and accuracy is to run the program making use of 48 nodes in the heated channel and a maximum Courant number as large as 0.9; however, stability maps obtained with 96 and 24 nodes will be also presented, in order to clearly point out the level of convergence reached in the addressed conditions.

The dimensionless equation of state adopted in the code represents an approximation of the mentioned unique relationship between dimensionless density and dimensionless enthalpy, $\rho^*(h^*)$, that was observed to hold for water at different supercritical pressures and, at a reasonable extent, also for different fluids.

3.2. Transient program in dimensional form (TRANSDIM)

The program, already used in Ambrosini (2009), is based on the semi-implicit discretisation of the original mass, momentum and energy balance equations, whose dimensionless form is at the basis of the previous described program. The equations are:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho w}{\partial z} = 0 \quad (12)$$

$$\frac{\partial \rho w}{\partial t} + \frac{\partial \rho w^2}{\partial z} + \frac{\partial p}{\partial z} = -\rho g - \left[\frac{f}{D_h} + K_{in} \delta_d(z) + K_{out} \delta_d(z-L) \right] \frac{\rho w^2}{2} \quad (13)$$

$$\frac{\partial \rho h}{\partial t} + \frac{\partial \rho h w}{\partial z} = q_0'' \frac{\Pi_h}{A} f_q(z) \quad (14)$$

The adopted numerical discretisation method is just the same as the one adopted for the linearised code, except for the fact that in this case a transient solution is obtained and stability is checked by simply running the code with a constant pressure drop at constant operating conditions (namely, pressure drop across the channel and inlet enthalpy) except for a continuous increase in the imposed heating power. This increase, kept very slow to avoid introducing unwanted transient effects, brings eventually the system to show oscillations, thus allowing to identify the corresponding stability threshold.

The critical value of the power-to-flow ratio observed at the threshold of instability is then translated into dimensionless form obtaining a corresponding value of N_{TPC} . With due caution in relation to the differences in the adopted boundary conditions, this value can be directly compared to the stability boundary obtained by the dimensionless program.

The equation of state required for closing the balance equations is introduced in the program in the form of very detailed tables of values of density as a function of enthalpy; a technique of direct access to the tables and a linear interpolation allow obtaining at the same time a reasonable computational effort in running the calculations and a good accuracy. Four different tables were generated from the NIST (2002) package for the fluids addressed in the problem (water, carbon dioxide, refrigerant R23 and ammonia) at the specific operating pressures listed in Table 1. As a consequence, in the calculations the variation of fluid properties as a function of the local value of pressure is not accounted for.

Also this code was used with 48 nodes, providing the basis for a direct comparison with the stability maps generated by the dimensionless code.

3.3. RELAP5/MOD3.3

RELAP5 (SCIENTECH Inc., 1999) is a well known system code adopted for the safety analysis of LWRs. In the previously mentioned works by the author, it was noted that the code does not show numerical problems when used for water at supercritical pressures, at least if no trans-critical condition is considered. On the contrary, it always showed reasonable agreement with the stability maps obtained by the dimensionless linearised code.

The main reason for using the code in this frame is to adopt a completely “diverse”, independent and verified tool whose results can be compared with those of the in-house codes. In fact, both these codes are derived by a similar root in terms of balance equations and discretization methods: an independent check was considered necessary.

Obviously, the use of the RELAP5 code is limited to water, since this is the single fluid of the ones addressed in this work to be available in RELAP5. The nodalization adopted in this case is also very simple, including two “time-dependent volumes”, used to impose the pressure drop, connected by two single-junctions to a pipe with 48 nodes. A very thin and highly conductive heat structure is introduced to simulate the conditions of imposed heat flux.

A maximum time step of 0.1 s was used in all the calculations, in which, anyway, the built-in Courant number limitation of the code is automatically applied to put an upper bound to the actually adopted time step. Also in the case of RELAP5, the strategy for achieving the stability threshold consisted in running calculations at constant boundary conditions with increasing the heating power up to the point in which flow oscillations are observed. The value of the power-to-flow ratio at the threshold of stability was then converted into a corresponding value of N_{TPC} , as in the case of the in-house transient program, to be compared with the stability boundary generated by the linearised code.

4. Obtained results

4.1. Stability thresholds for the vertical channel with water

Figs. 2 and 3 report the results obtained by the linear stability program for the vertical channel cases. In particular, Figs. 2 and 3 report information on the distributions of z_R and of the ratio between the fluid transit time and the period of oscillations, T_{trans}/T .

It must be explicitly noted that in these analyses the value of the friction parameter, f , that must be explicitly specified in the linearised program, is assumed constant and equal to the value that it is assumed to approximately hold in the addressed conditions; since the friction factor is a function of the Reynolds number, this represents an approximation introduced in the calculation, though the Reynolds number in the addressed cases is large enough to be in a region in which the friction factor mostly depends on pipe roughness. Moreover, the adopted value 0.03 for the Froude number for vertical channel corresponds to an inlet velocity equal to $w_{in} = \sqrt{Fr g L} \approx 1.12$ m/s with the geometrical data of the present problem, which is not exactly the case in the transient calculations to be compared with the map.

Anyway, as shown in Figs. 4 and 5, the latter aspect is not expected to pose relevant problems in establishing the stability threshold in terms of N_{TPC} , though it may change the value of the ratio T_{trans}/T ; this has the consequence that while the comparison of the stability margin from the maps with the data of the transient programs is fully meaningful, the comparison of the values of

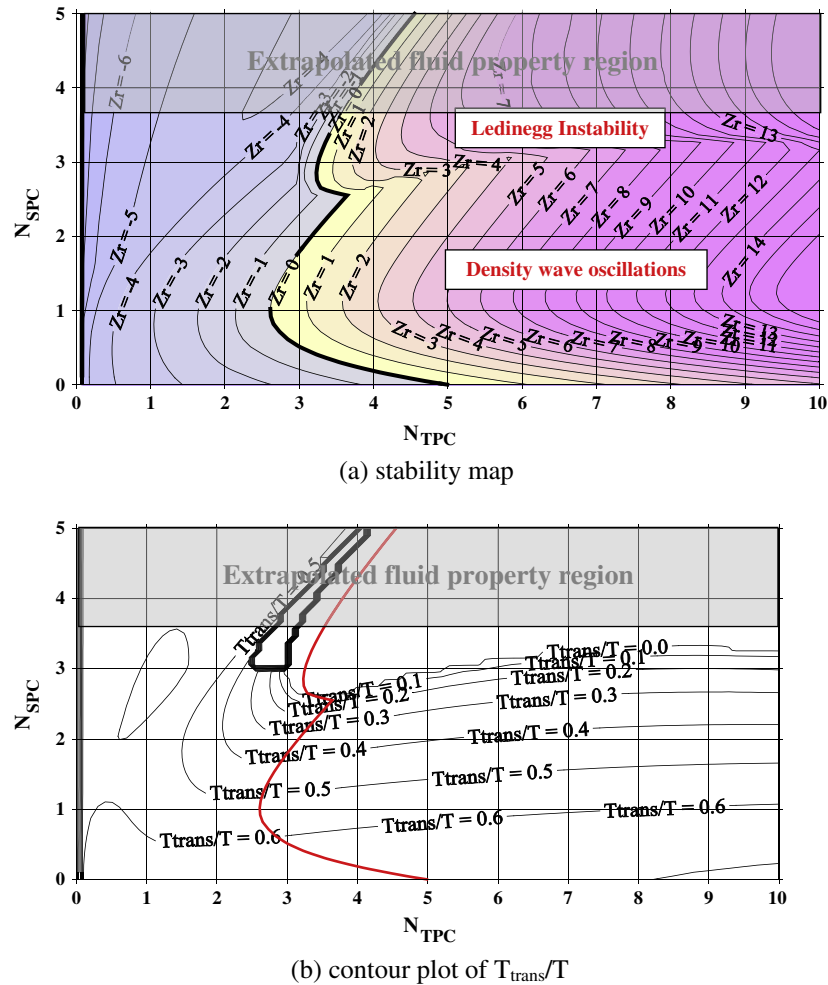


Fig. 2. Reference stability map and contour plot of T_{trans}/T for the case of vertical channel with $K_{in} = K_{out} = 20$ ($A = 7$, $Fr = 0.03$, $C_{max} = 0.9$, $N_{nodes} = 48$).

T_{trans}/T will be affected by some degree of indeterminacy. In fact, as noted in the figures, the effect of changing the Froude number in the considered range (10^3 stands for a nearly horizontal channel) is mainly to shift the region where the period of oscillations becomes infinite, identifying the occurrence of an excursive type of instability, i.e. the Ledinegg one. The discussion will go further in depth on these aspects later on.

Fig. 6 presents some of the time trends obtained by RELAP5/MOD3.3 for N_{TPC} , basing on the calculated values of heating power and flow rate. As it can be noted, the threshold values of N_{TPC} at the onset of instability are identified by a visual criterion that is reliable enough when the rate of channel power increase is slow. In this respect, an effort was made in this work to reduce as feasible as possible this rate in order to approach instabilities in a quasi-steady situation; in fact, part of the general inaccuracy already observed in the comparison between stability maps and RELAP5 stability thresholds was due to the methodology adopted for threshold detection (compare with the results in Ambrosini and Sharabi (2007)).

Fig. 7 presents an overall view of the results obtained for water at 25 MPa and different outlet throttling by RELAP5 and by the linearised dimensionless in-house code. Some features appearing in the figure are worth to be pointed out; in particular:

- a very good agreement is obtained between the continuous stability boundaries and the discrete threshold conditions evaluated by RELAP5;

- the effect of outlet throttling on stability is clearly observed and is at a very good extent predicted to be the same by both models;
- the slight difference between the solid lines, obtained by the linearised program with 48 nodes, and the dashed lines, obtained with 96 nodes, points out some effect of numerical diffusion: this aspect will be addressed in greater detail later on;
- taking into account the nearly exponential excursion in the curve reported in Fig. 6 at $N_{SPC} = 3.0$, it can be clearly understood that in such case it is rather difficult to identify the threshold value of N_{TPC} at which the excursion really start; this is not the case of the oscillatory behaviour observed at lower pseudo-subcooling, that is more easily discriminated from the slower drift of N_{TPC} occurring during channel power increase.

In summary, the obtained results show a very good agreement between RELAP5 and the linearised in-house code in terms of the threshold value of N_{TPC} . On the other hand, comparing the corresponding results obtained for T_{trans}/T reported in Table 2 with the values of the contour plots of this variable reported in the previous Figs. 2b and 3b it can be noted that:

- the two models agree in predicting the range of variation of this parameter from roughly 0.6–0.7 down to zero at increasing pseudo-subcooling;
- the quantitative agreement is only approximate, though quite good in many instances.

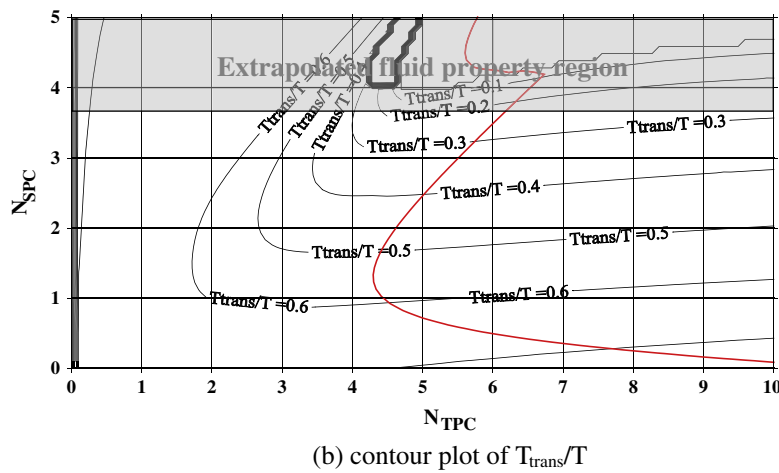
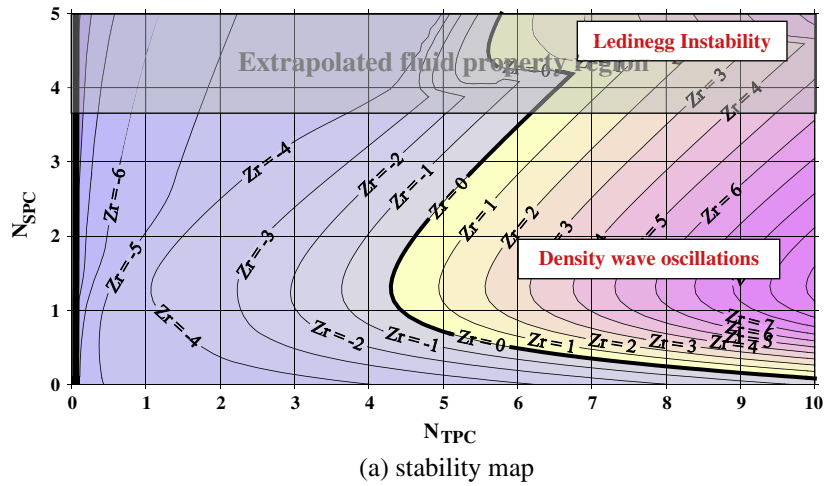


Fig. 3. Reference stability map and contour plot of T_{trans}/T for the case of vertical channel with $K_{in} = 20$ and $K_{out} = 2$ ($A = 7$, $Fr = 0.03$, $C_{max} = 0.9$, $N_{nodes} = 48$).

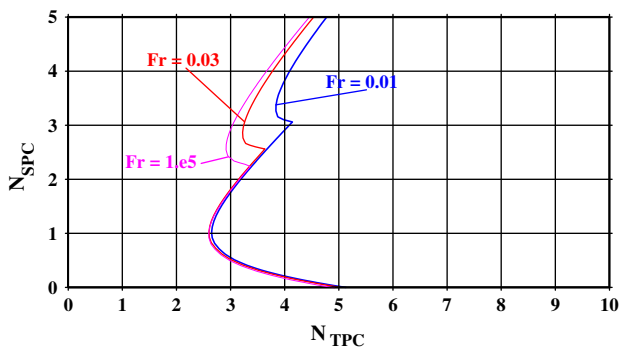


Fig. 4. Effect of the Froude number on the stability boundary in the case with $K_{in} = K_{out} = 20$ ($A = 7$, $C_{max} = 0.9$, $N_{nodes} = 48$).

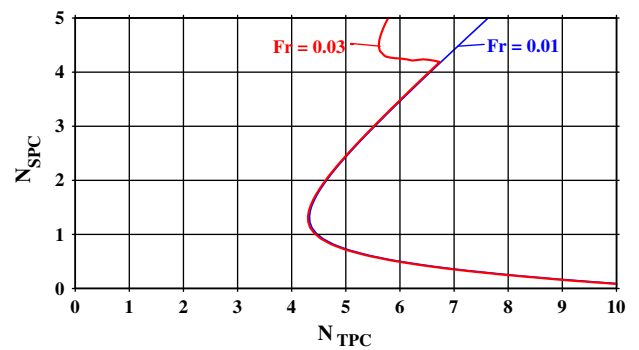


Fig. 5. Effect of the Froude number on the stability boundary in the case with $K_{in} = 20$ and $K_{out} = 2$ ($A = 7$, $C_{max} = 0.9$, $N_{nodes} = 48$).

Among the possible items contributing to the limited accuracy of the latter comparison, the following ones can be mentioned:

- the ratio T_{trans}/T was evaluated by computing the fluid transit time through the channel on the basis of the nodal velocities and the period of oscillations was guessed by graphical means counting the oscillations observed in N_{TPC} during time: both these techniques are obviously affected by some uncertainty;

- as clearly seen from Figs. 2b and 3b, the region for excursive behaviour is bounding from above the region of density-wave oscillations, where T_{trans}/T is non-zero; on the other hand, since Figs. 4 and 5 have shown that a change in the Froude number has mainly the effect to shift the location of the region of Ledinegg instability, the variation of the Froude number in RELAP5 calculations makes only qualitative the comparison of the related values of T_{trans}/T with the ones obtained in the map at $Fr = 0.03$.

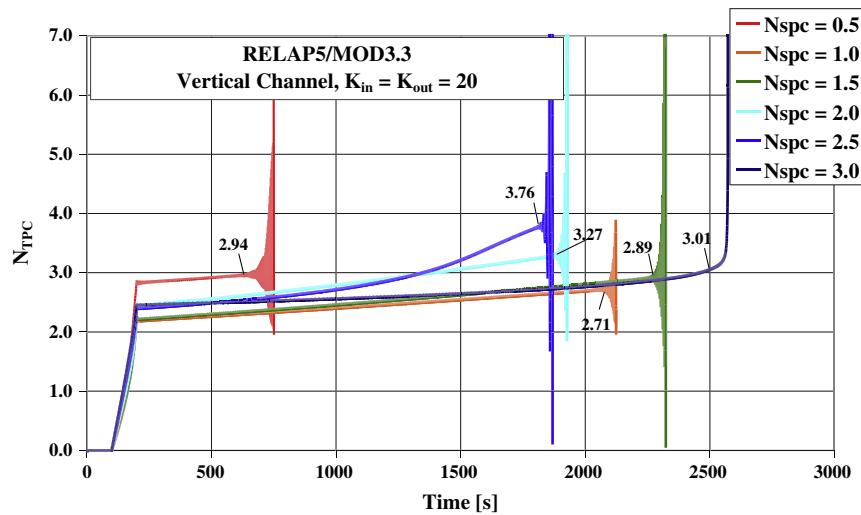


Fig. 6. RELAP5/MOD3.3: time trends of N_{TPC} for the cases with vertical channel and $K_{in} = K_{out} = 20$ at different inlet enthalpies.

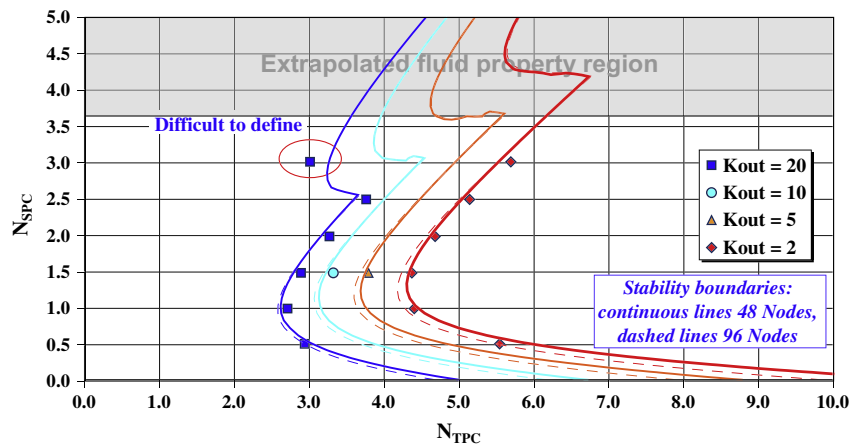


Fig. 7. Comparison of RELAP5/MOD3.3 data for vertical channel and water with the stability boundaries obtained with $A = 7$, $Fr = 0.03$, $C_{max} = 0.9$.

Table 2

Ratio of the fluid transit time in the channel to the period of oscillations at the onset of instability as evaluated by RELAP5/MOD3.3 for the vertical channel cases with water.

N_{SPC}	$K_{out} = 20$	$K_{out} = 10$	$K_{out} = 5$	$K_{out} = 2$
0.51	0.63	–	–	0.71
1.00	0.53	–	–	0.60
1.49	0.47	0.48	0.51	0.52
1.99	0.35	–	–	0.46
2.50	0.22	–	–	0.40
3.01	0.00	–	–	0.34

4.2. Stability thresholds for vertical and horizontal channels with different fluids

The TRANSDIM program was then applied to the cases of the vertical and horizontal channel with $K_{in} = K_{out} = 20$ at different values of the pseudo-subcooling. As already mentioned, only a subset of conditions was explored with carbon dioxide, ammonia and refrigerant R23 with respect to the set of conditions considered for water, partly due to the different field of existence of the fluids. The stability thresholds in terms of N_{TPC} were determined as already shown in the case of RELAP5 in Fig. 6, i.e., by detecting the onset of instability by visual means in the time trend of N_{TPC} . Again,

while in the case of oscillatory behaviour this threshold could be detected rather accurately, in the case of excursive behaviour the already mentioned indeterminacy comes into play; as a consequence, the discrepancies observed in this regard can be quite easily explained.

Figs. 8 and 9 show that the stability thresholds obtained for the different fluids are reasonably crowded along the same boundary stability lines obtained by the linearised dimensionless program; Table 3 allows to compare more quantitatively the obtained data. This result can be considered a trivial consequence of the already observed relative independence of the dimensionless density as a function of dimensionless enthalpy in the presently adopted formulations; however, performing independent calculations by solving the balance equation in dimensional form for the different fluids provides a confirmation that other aspects (e.g., the limited differences in friction factors due to the differences in the Reynolds numbers) play a minor role. With respect to the findings previously obtained in this regard for the smooth pipe without throttling considered in Ambrosini (2009), the present data represent a more convincing demonstration of the considerable degree of universality that can be obtained by the dimensionless definitions proposed at the beginning of this research.

It is also interesting noting that the comparison of vertical versus horizontal channel data shows a substantial coherence between the linear stability boundaries and the thresholds obtained

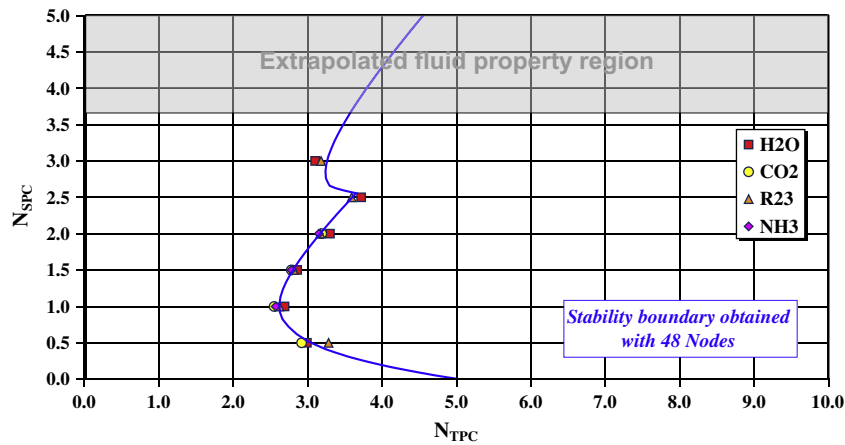


Fig. 8. Comparison of TRANSDIM data for vertical channel and the fluid-to-fluid comparison with the stability boundaries obtained with $A = 7$, $Fr = 0.03$, $C_{max} = 0.9$.

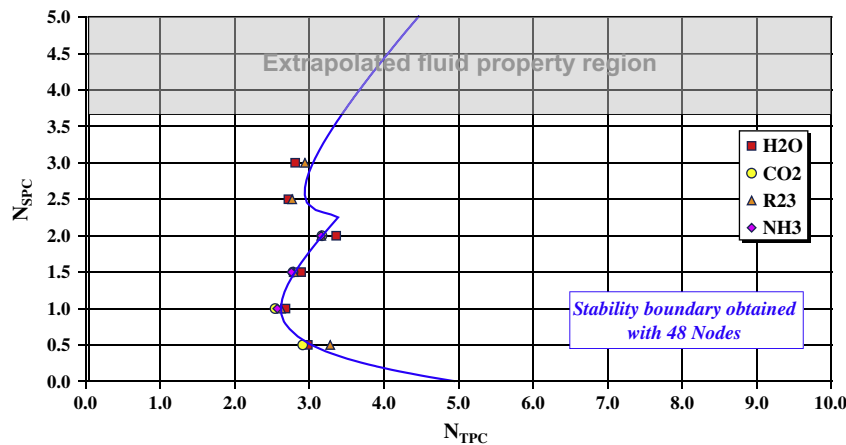


Fig. 9. Comparison of TRANSDIM data for horizontal channel and the fluid-to-fluid comparison with the stability boundaries obtained with $A = 7$, $Fr = 10^5$, $C_{max} = 0.9$.

Table 3

Values of the stability threshold in terms of N_{TPC} at the onset of instability as evaluated by TRANSDIM for $K_{in} = K_{out} = 20$ with different fluids.

Channel orientation	N_{SPC}	Water	CO ₂	R23	NH ₃
		$N_{TPC,threshold}$	$N_{TPC,threshold}$	$N_{TPC,threshold}$	$N_{TPC,threshold}$
Vertical	0.5	2.99	2.92	3.28	–
Vertical	1	2.69	2.55	2.62	2.57
Vertical	1.5	2.86	2.78	2.81	2.78
Vertical	2	3.3	3.19	3.17	3.15
Vertical	2.5	3.72	–	3.59	–
Vertical	3	3.1	–	3.18	–
Horizontal	0.5	2.98	2.91	3.28	–
Horizontal	1	2.68	2.54	2.62	2.57
Horizontal	1.5	2.89	2.78	2.79	2.76
Horizontal	2	3.36	3.17	3.16	3.16
Horizontal	2.5	2.72	–	2.77	–
Horizontal	3	2.81	–	2.94	–

at variable pseudo-subcooling by the TRANSDIM program; in particular, it is evident that the linearised program and the transient one show the same behaviour in relation to the change in the character of instability observed for the points at $N_{SPC} = 2.5$, being oscillatory for the vertical channel and excursive for the horizontal channel.

Finally, the comparison of the data obtained by TRANSDIM for the ratio of the transit time to the period of oscillations at the onset of instability, reported in Table 4, with the contour plots shown in

Table 4

Ratio of the fluid transit time in the channel to the period of oscillations at the onset of instability as evaluated by TRANSDIM for $K_{in} = K_{out} = 20$ with different fluids.

Channel orientation	N_{SPC}	Water	CO ₂	R23	NH ₃
Vertical	0.5	0.61	0.64	0.68	–
Vertical	1	0.52	0.53	0.56	0.53
Vertical	1.5	0.48	0.44	0.48	0.43
Vertical	2	0.35	0.34	0.40	0.32
Vertical	2.5	0.21	–	0.31	–
Vertical	3	0.00	–	0.00	–
Horizontal	0.5	0.61	0.63	0.68	–
Horizontal	1	0.51	0.52	0.55	0.53
Horizontal	1.5	0.40	0.41	0.45	0.42
Horizontal	2	0.28	0.30	0.35	0.38
Horizontal	2.5	0.00	–	0.00	–
Horizontal	3	0.00	–	0.00	–

Figs. 2b and 10b leads to conclusions similar to those already discussed for the cases of the vertical channel with water. In particular, the general trends are confirmed, while the quantitative comparison seems to be affected by different sources of inaccuracy owing to the different boundary conditions used in the calculations, mainly in terms of the Froude number.

4.3. Effect of numerical diffusion

Fig. 11 shows the stability boundaries obtained by the linearised model with different numbers of nodes, with a maximum

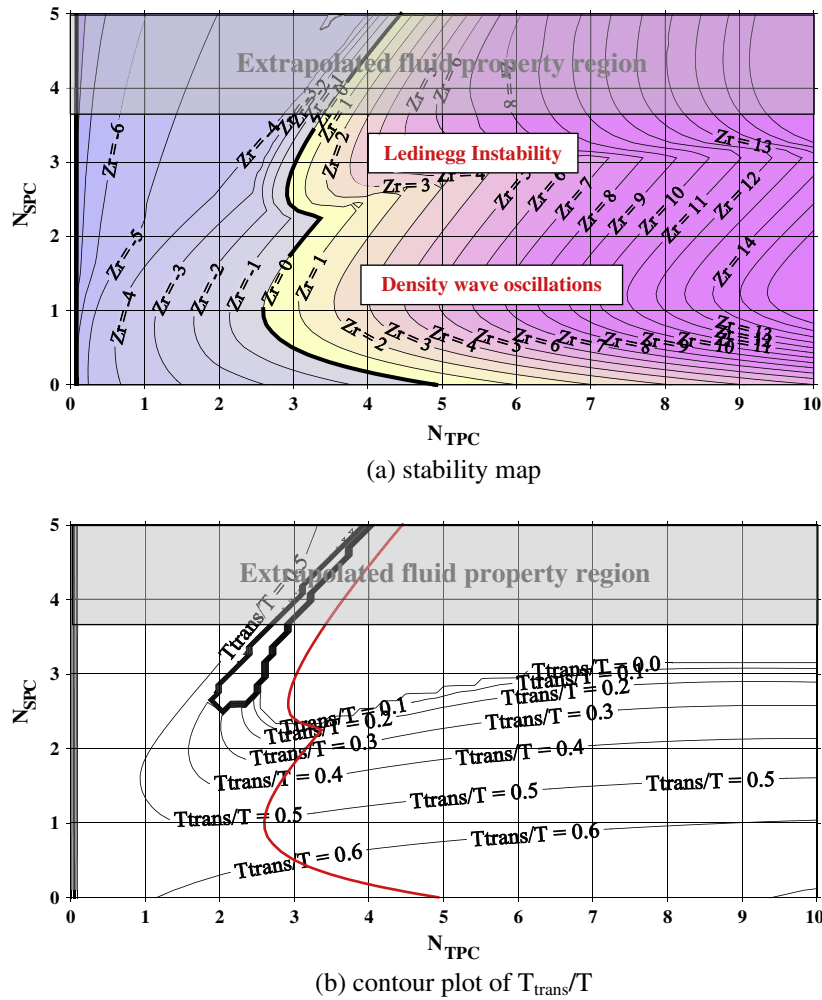


Fig. 10. Reference stability map and contour plot of T_{trans}/T for the case of horizontal channel with $K_{in} = K_{out} = 20$ ($A = 7$, $Fr = 10^5$, $C_{max} = 0.9$, $N_{nodes} = 48$).

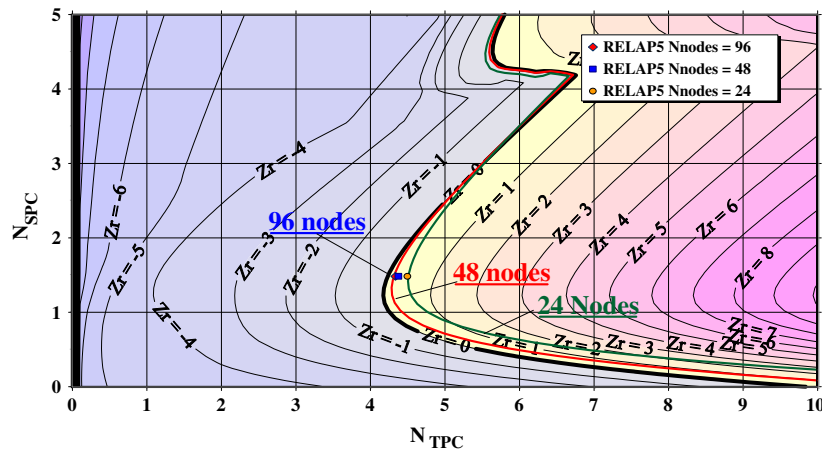


Fig. 11. Effect of the number of axial nodes on stability prediction for the vertical channel with $K_{in} = 20$, $K_{out} = 2$ and water ($A = 7$, $Fr = 0.03$, $C_{max} = 0.9$).

Courant number equal to 0.9. The results obtained for the stability threshold by RELAP5 for the case with $N_{SPC} = 1.5$ code are also represented by markers. As it can be noted, the two programs, being both affected by a first order truncation error, show that using 24, 48 or 96 nodes affects the stability thresholds as expected from

basic considerations about the effects of discretisation. The behaviour of the system code and the in-house one in this respect is similar, though not exactly equal also because of the different maximum time steps adopted, being 0.1 s for RELAP5 and adapted to the maximum Courant in the case of the linearised code. In this

regard, an additional difference that must be taken into account is also the possibly different upper limits imposed in the two programs on the maximum Courant numbers.

Additional information on this aspect can be drawn by observing the difference between the solid and the dashed stability boundaries reported in Fig. 7, obtained respectively with 48 and 96 nodes. As it can be noted, the effect of numerical diffusion is seen to increase with N_{TPC} and decrease with N_{SPC} . A possible reason for this behaviour may be related to the larger differences in local velocities along the channel occurring at high power-to-flow ratio and low pseudo-subcooling, that makes the Courant number to strongly vary along the channel; as a consequence, imposing a large maximum value of the Courant number results anyway in having also nodes in which the Courant number will be small enough to enhance diffusion.

The above confirms that it is always worth checking the level of convergence in the calculations, whenever it is necessary to achieve a high level of accuracy in predicting stability thresholds. Indeed, the variability of fluid velocity along the channel does not allow assuring high accuracy by just imposing a large maximum value of the Courant number.

4.4. Effect of pressure drop across the channel on stability

The calculations performed with water and the other fluids by the TRANSDIM program were performed initialising the steady-state conditions at some reasonable but arbitrary value of flow rate and then increasing the heating power up to the point at which unstable behaviour was observed. This procedure introduces some degree of indeterminacy in the pressure drop across the channel in the different operating conditions and, as a consequence, on the Froude number whose value, w_{in}^2/gL , is obtained by varying inlet velocity. On the other hand, in the case of RELAP5, a constant value of the pressure drop was mostly used (around 0.12 MPa) in the analyses at different pseudo-subcooling and outlet throttling: this has obviously again an impact on the value of Froude number which varies in each transient calculation, while it is instead kept constant in the stability maps obtained by the linearised code.

The good comparison obtained between the stability thresholds evaluated by the transient programs (RELAP5 and TRANSDIM) and the stability boundaries obtained by the linearised code suggests that the parameters influencing the Froude number (inlet velocity and, as a consequence, the channel overall pressure drop) have a limited effect on the stability prediction in the addressed range

of the other parameters. This aspect was checked by running RELAP5 calculations at different values of the overall channel pressure drop for two relevant cases at different outlet throttling; the considered values of the overall pressure drop were 0.12 MPa and 0.22 MPa respectively; the results of these calculations are shown in Figs. 12 and 13 showing that, notwithstanding the different power and flow rate levels obtained with the different pressure drops, the thresholds values of N_{TPC} at the onset of stability are nearly the same. This confirms the observation of a minor role of the Froude number in these operating conditions, already noted considering Figs. 4 and 5.

4.5. Density-wave and Ledinegg instability mechanisms

The contour plots reported for the ratio of the transit time to the period of oscillations as a function of N_{TPC} and N_{SPC} (see e.g., Figs. 2b, 3b and 10b) show that this ratio has a continuous variation in the region of density-wave oscillations approaching zero where the Ledinegg instability region is reached. This continuity between the two instability phenomena, already observed in the case of boiling channels (Ambrosini et al., 2000), is presently confirmed also by the results obtained by RELAP5 and TRANSDIM for water and the other considered fluids (Tables 2 and 4).

Fig. 14 helps in understanding the reason for the increase in the period of oscillations observed at increasing values of N_{SPC} for a particular case; in fact, it can be seen that the two maximum eigenvalues characterising the dynamics of the system, as predicted by the linearised program, slowly drift from a complex conjugated pair with decreasing imaginary part to a couple of real roots, parting from each other along the real axis. Since the imaginary part of the complex eigenvalues is directly related to the frequency of oscillations, it is understood that, with increasing N_{SPC} along the arrow reported in the contour plot on the right of Fig. 14, longer and longer periods of oscillations are obtained, until the oscillations degenerate into an excursive behaviour (i.e., with infinite period).

In this respect, it is interesting to mention that this picture, obtained by the analysis of computed results, is fully coherent with the basic understanding about density-wave and Ledinegg instability. Both of them, in fact, refer to situations in which the channel shows a response to flow perturbations opposite than in usual cases, i.e. characterised by a decrease in the internal pressure drop for increasing flow rate and vice versa. When this occurs for some non-zero frequency of the flow perturbations, i.e. in dynamic conditions, density-wave oscillations are observed; on the other hand,

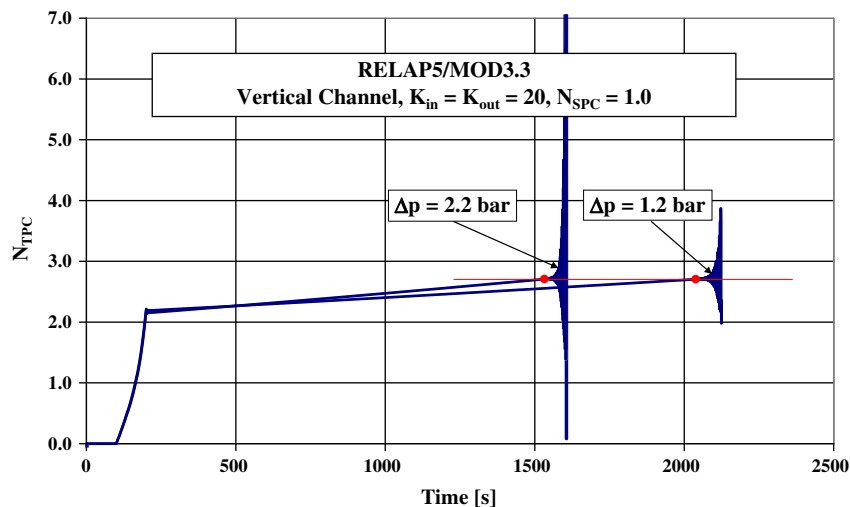


Fig. 12. Effect of pressure drop on RELAP5 stability prediction for the vertical channel with water and $K_{in} = 20$, $K_{out} = 20$, $N_{SPC} = 1.0$.

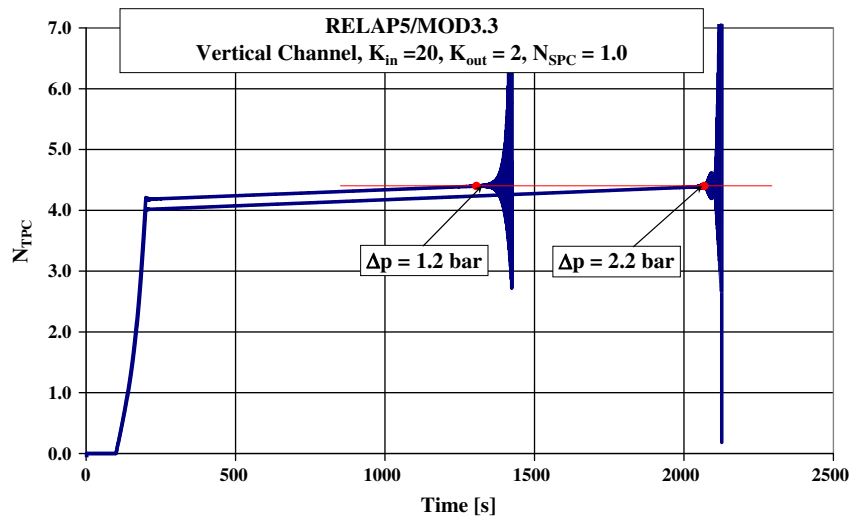


Fig. 13. Effect of pressure drop on RELAP5 stability prediction for the vertical channel with water and $K_{in} = 20$, $K_{out} = 2$, $N_{SPC} = 1.0$.

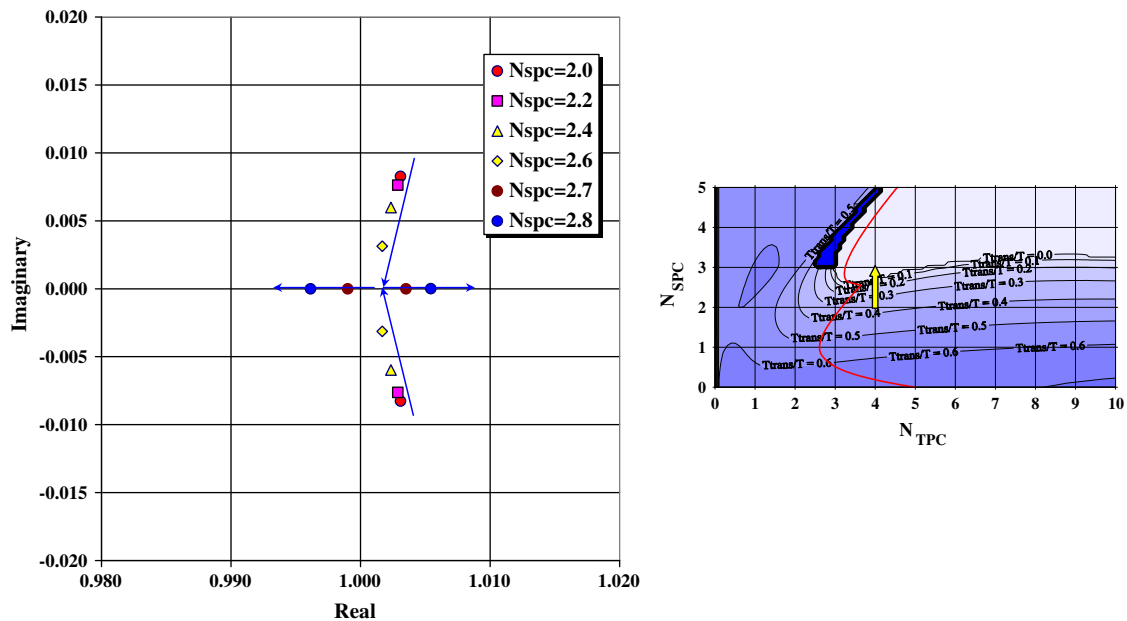


Fig. 14. Change in the two largest magnitude eigenvalues with increasing N_{SPC} at $N_{TPC} = 4.0$ in the case of the vertical channel with $K_{in} = K_{out} = 20$ ($A = 7$, $Fr = 0.03$, $C_{max} = 0.9$, $N_{nodes} = 48$).

when this occurs at zero frequency, i.e. in static conditions, the Ledinegg instability occurs.

5. Conclusions

The additional results presented in this paper about the simple heated channel containing fluids at supercritical pressure demonstrated more clearly some facts, already implied in what suggested by Ambrosini and Sharabi (2006, 2008), that were given specific attention in further analyses developed up to now.

In particular:

- the stability maps obtained in dimensionless form by the adopted in-house linearised program are confirmed by the prediction of rather different tools and can be helpful in understanding the behaviour obtained by system codes;
- the fact that a specific channel with arbitrary geometrical features can be predicted by completely diverse codes to behave

in close agreement with these stability maps confirms the generality of the adopted dimensionless groups;

- in addition, the fact that the use of rather different fluids, as those considered in the present work, provides nearly the same stability thresholds for the addressed system represents a demonstration of the reasonable degree of universality that the proposed set of dimensionless numbers possesses; this conclusion is presently limited to conditions of imposed heating rate, which are anyway of overwhelming importance for nuclear reactor core cooling;
- the methodology adopted in the analysis, as shown in previous works related to the stability of different flow systems, has the interesting by-product of clearly pointing out the effects of numerical discretization, thus representing a useful tool to provide guidelines in the use of system codes;
- the combination of linear stability and transient calculations allowed understanding the effect of the overall pressure drop

across the channel on the stability boundary, showing a very weak effect of this parameter in some of the considered operating conditions;

- a basic continuity between the static Ledinegg instability and the dynamic density-wave oscillation mechanisms is observed by the contour plots of the ratio between the fluid transit time and the period of oscillations.

On the basis of the understanding gained in relation to the simple heated channel with imposed heat flux and forced flow, work is presently in progress to extend the above considerations to systems in which heating is controlled by temperature and flow is due to natural circulation; the results of this work will be reported in forthcoming papers.

Acknowledgements

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